

Module 7: Molecular Genetics - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. In DNA, the base adenine always pairs with:
2. A. cytosine
3. B. guanine
4. C. thymine
5. D. another adenine
6. Answer: C

7. Why: The base-pairing rules are A with T and C with G. This complementarity lets DNA be copied accurately.

8. DNA replication is described as "semiconservative" because each new double helix contains:
9. A. one original strand and one new strand
10. B. two brand-new strands
11. C. two original strands
12. D. no strands at all
13. Answer: A

14. Why: Each daughter molecule keeps one parental strand as a template, so half of the original is conserved.

15. Which sequence shows the normal flow of genetic information?
16. A. protein, then RNA, then DNA
17. B. RNA, then DNA, then protein
18. C. protein, then DNA, then RNA
19. D. DNA, then RNA, then protein
20. Answer: D

21. Why: Transcription copies DNA into RNA, and translation reads codons to build protein: DNA to RNA to protein.
22. A mutation changes a single DNA base. Why might this matter?
23. A. DNA cannot change, so nothing happens.
24. B. It can change a codon, which may change an amino acid and the protein.
25. C. It always improves the organism.
26. D. It affects only RNA and never proteins.
27. Answer: B
28. Why: A base change can alter a codon and therefore the protein. Effects range from none to significant.